MILITARY SPECIFICATION

CABLE ASSEMBLY: T4016-4, W14F

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of cable assembly, entitled Cable Assembly: T4016-4, W14F which is used in the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8808362	- Container, Package, For CONUS Shipments.
8821506	- Cable Assembly, T4016-4, W14-F
8863983	- Carton, Packing, Ammunition, for Chassis Assembly.
EL-8821506	- Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

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3.2 Assembly. The cable assembly shall comply with all requirements specified on the drawing (dwg) 8821506, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Soldering. All soldering shall comply with the requirements of MIL-STD-1460.

3.6 Continuity. The cable assemblies shall exhibit point to point continuity in accordance with Table I when tested as specified in 4.4.1.

Table I
Continuity

Arm Position		Fuze Position	
From	To	From	To
P1-S	P6-A	P1-S	P6-M
P1-R	P6-B	P1-R	P7-J
P1-T	P6-C	P1-T	P7-K
P1-V	P6-g	P1-V	P7-L
P1-E	P1-D	P1-A	P6-J
P1-K	P1-C	P1-B	P6-K
P1-F	P1-G	P1-U	P6-L
P1-H	P1-J	P1-W	P7-M
P1-A	P7-A	P1-D	P1-E
P1-B	P7-B	P1-C	P1-K
P1-U	P7-C	P1-a	P7-f
P1-W	P7-g		
P1-L	P7-M		
P1-a	P7-b		
P7-b	P7-R		
P7-R	P1-Z, P6-f		

3.7 Insulation resistance. The insulation resistance between electrically isolated wires and between ungrounded wires and the connector shells shall be not less than 100 megohms when a direct current (dc) potential of 500 plus or minus (±) 50 volts (v) dc is applied. The current shall be limited to 5 milliamperes (ma). Testing shall be specified in 4.4.2.

3.8 Resistance.

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3.8.1 Contact. The resistance of the circuit involving relay contacts between any two contacts in the make position, when measured at the connector terminals, shall not exceed 0.02 ohms when tested as specified in 4.4.3.1.

3.8.2 Circuit. The circuit resistance from P7-R to P7-f shall be 75 ± 3.75 ohms and the circuit resistance from P1-a to P7-b, with the switch in the arm position, shall be 125 ± 6.25 ohms when tested as specified in 4.4.3.2.

3.9 Environmental testing (see 4.4.4).

3.9.1 High temperature. The cable assembly shall show no evidence of deterioration such as cracking, swelling, loosening of insulation, distortion, broken parts, or other damage when tested as specified in 4.4.4.1. After this test, the cable assembly shall comply with the requirements of 3.7.

3.9.2 Low temperature. The cable assembly shall show no evidence of deterioration such as distortion, cracking, swelling, loosening of insulation, embrittlement, or other damage when tested as specified in 4.4.4.2. After this test, the cable assembly shall comply with the requirements of 3.7.

3.9.3 Vibration. The cable assembly shall show no evidence of electrical impairment or of mechanical damage such as visible loosening or deformation of any part of the switch or change in switch position when tested as specified in 4.4.4.3. After this test, the cable assembly shall comply with the requirements of 3.6 and 3.8.

3.10 Workmanship.

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.10.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8821506) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests need not be conducted in order specified in Table II.

Table II
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Continuity (see 3.6)		X
3. Insulation resistance (see 3.7)		X
4. Resistance (see 3.8)		X
5. High temperature (see 3.9.1)		X
6. Low temperature (see 3.9.2)		X
7. Vibration (see 3.9.3)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

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4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table III and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table III
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table IV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Cable assembly prior to closing switch box (see dwg 8821506).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Solder improper (see 3.5)	Visual
102.	Cracked or torn insulation	Visual
*103.	Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to adjacent pins)	Visual/Gage
**104.	Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/16 inch in depth relative to the adjacent sockets)	Visual/Gage
105.	Any component or hardware missing or loose	Visual/Manual
Minor:		
201.	Evidence of poor workmanship (see 3.10)	Visual
202.	Serialization not in accordance with 3.4	Visual

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*In the event pin(s) do not appear to be within 1/16 inch in height relative to the adjacent pins, Gage (see 4.3.4) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.063 inch max. from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface of all measurements of pin heights.

**In the event socket inserts do not appear to be within 1/16 inch in depth relative to the adjacent socket, Gage (see 4.3.4) shall be used to determine the variation from the normal depth. Reject any insert that varies by more than 0.063 inch max. from the normal socket insert depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward edge of the connector insert shall be utilized as the reference surface for all measurements of socket depths.

4.3.2.1.2 Cable assembly, T4016-4, W14F (see dwg 8821506).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Length from base of switch box to end of connector plugs, minimum	Gage
102.	Connector pins improperly positioned	Gage
103.	Connector pins cracked or bent	Visual
104.	Connector inserts cracked	Visual
105.	Cracked, torn, or damaged insulation sleeving	Visual
106.	Marking missing, improper or unidentifiable	Visual
107.	Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.10)	Visual

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4.3.2.1.3 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Packaging, packing, and marking not in accordance with dwgs 8808362 and 8863983	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.3 Testing. All tests shall be performed. Tests need not be performed in the order specified below:

Test	Test Classification
a. Continuity	Major
b. Contact resistance	Major
c. Insulation resistance	Major
d. Vibration	Major

Of these tests, d is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Continuity, contact resistance, and insulation resistance. Each cable assembly shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.3.1. If any item fails to comply with 3.6, 3.7, and 3.8.1, the item shall be rejected and removed from the lot.

4.3.3.2 Vibration. One sample cable assembly shall be selected from each 50 cable assemblies, or portion thereof produced and tested in accordance with 4.4.4.3. If the sample fails to comply with 3.9.3, the 50 cable assemblies shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8821506). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

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a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Continuity. The continuity between the points indicated in Table I shall be determined by utilizing an ohmmeter or other suitable tester.

4.4.2 Insulation resistance. A potential of 500 ± 50 vdc shall be applied in accordance with Table V. The potential shall be applied for a duration of not less than 15 seconds. The resistance shall be checked by means of an insulation resistance tester from which a direct reading in megohms can be obtained.

Table V

Insulation resistance
Switch position: Arm

Pin Energized	All other pins to ground except (including unused pins and shells)
P1-S	P6-A
P1-R	P6-B
P1-T	P6-C
P1-V	P6-g
P1-E	P1-D
P1-K	P1-C
P1-F	P1-G
P1-H	P1-J
P1-A	P7-A
P1-B	P7-B
P1-U	P7-C
P1-W	P1-g
P1-L	P1-M
P1-Z	P6-f, P1-a, P7-R, P7-f

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Switch position: Fuze

P1-S	P6-M
P1-R	P7-J
P1-T	P7-K
P1-V	P1-L
P1-A	P6-J
P1-B	P6-K
P1-U	P6-L
P1-W	P7-M
P1-a	P7-f, P7-R, P1-Z, P6-f

4.4.3 Resistance.

4.4.3.1 Contact. The contact resistance of the cable assemblies shall be checked in accordance with Table VI by the voltmeter-ammeter four-wire method.

Table VI

Contact resistance

From	To
P1-K	P1-C
P1-E	P1-D
P1-F	P1-G
P1-J	P1-H

4.4.3.2 Circuit. The circuit resistance of the cable assemblies shall be determined by the voltmeter-ammeter method.

4.4.4 Environmental testing (see 3.9).

4.4.4.1 High temperature. The cable assembly shall be placed in a test chamber and the test chamber temperature raised to and stabilized (see 6.4) at 165 degrees (°) ± 5° Fahrenheit (F) for a minimum of 2 hours. The cable assembly shall then be returned to ambient room temperature. No forced cooling shall be used in returning the cable assembly to ambient room temperature. After completion of this test, the cable assembly shall be tested as specified in 4.4.2.

4.4.4.2 Low temperature. The cable assembly shall be tested as specified in 4.4.4.1 except that the test chamber temperature shall be lowered to -65 ± 2°F. No forced warming shall be used in returning the cable assembly to ambient room temperature. After completion of this test, the cable assembly shall be tested as specified in 4.4.2.

4.4.4.3 Vibration. The cable assembly shall be rigidly mounted on a vibration platform. A simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion) shall be applied, the frequency being varied uniformly between the approximate limits of 10 and 55 Hertz (Hz).

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The entire frequency range from 10 to 55 to 10 Hz shall be transversed in approximately 1 minute. The motion shall be continuously applied for 15 minutes in each of three mutually perpendicular directions, one of which shall be parallel to the shaft of the switch. After completion of this test, the cable assembly shall be tested as specified in 4.4.1 and 4.4.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The cable assembly shall be packaged in accordance with dwg 8863983 using the specified moisture barrier.

5.1.2 Level C. The cable assembly shall be packaged in accordance with dwg 8863983 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The cable assembly shall be packed in accordance with dwg 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwgs 8808362 and 8863983 as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A094

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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